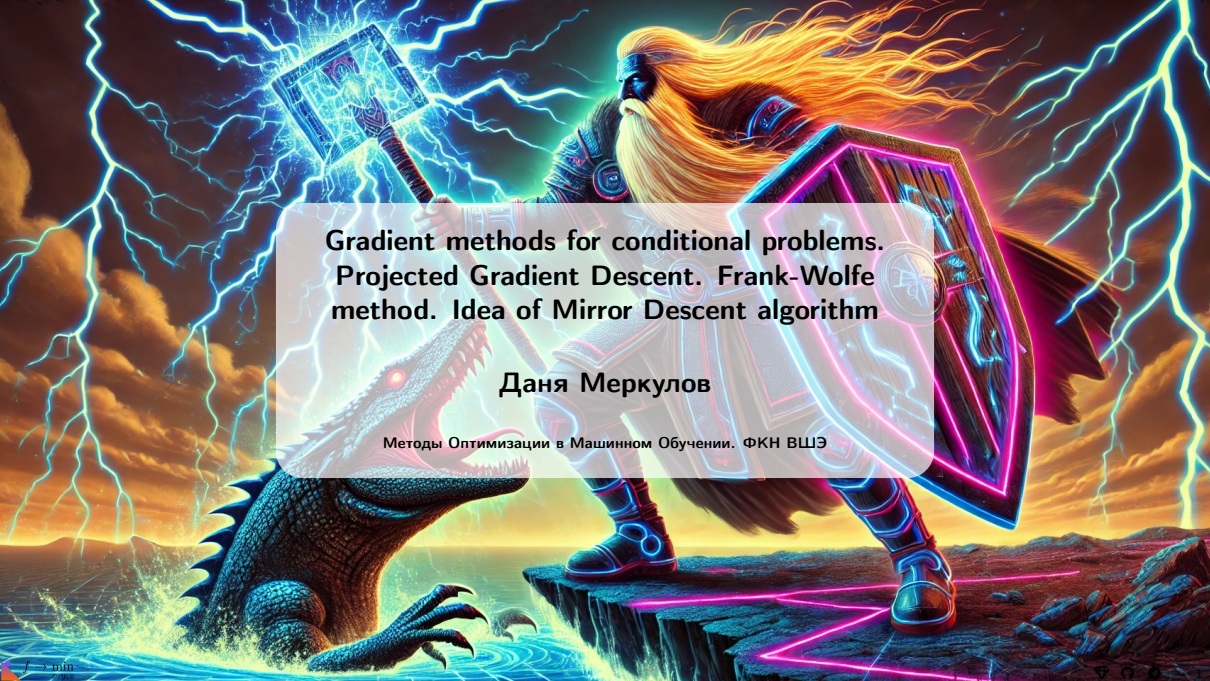




Gradient methods for conditional problems.  
Projected Gradient Descent. Frank-Wolfe  
method. Idea of Mirror Descent algorithm

Даня Меркулов

Методы Оптимизации в Машинном Обучении. ФКН ВШЭ

## Conditional methods

# Constrained optimization

## Unconstrained optimization

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- Any point  $x_0 \in \mathbb{R}^n$  is feasible and could be a solution.

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Is it possible to tune GD to fit constrained problem?

**Yes.** We need to use projections to ensure feasibility on every iteration.

## Example: White-box Adversarial Attacks

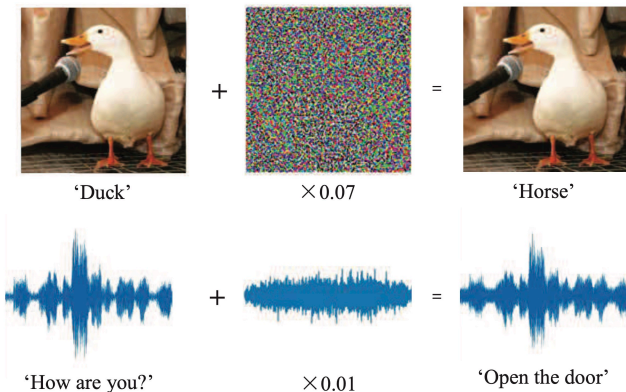


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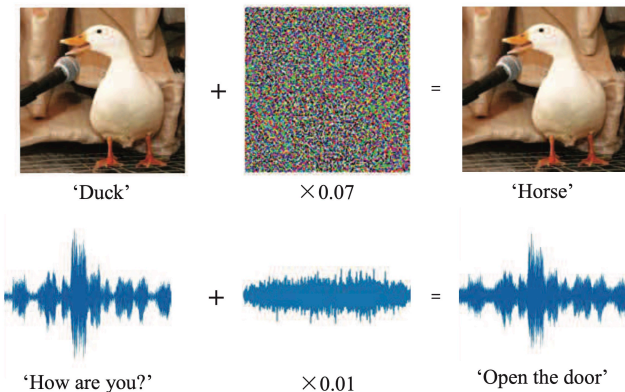
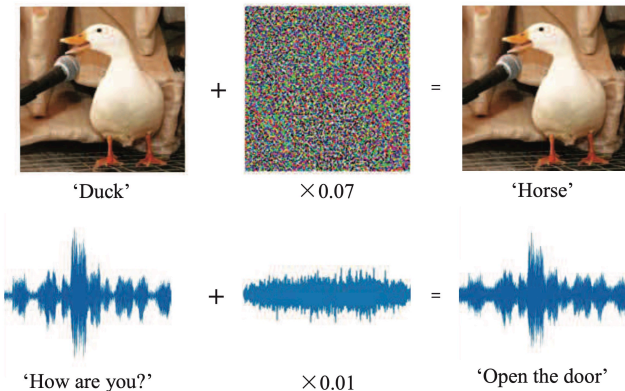


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- Mathematically, a neural network is a function  $f(w; x)$
- Typically, input  $x$  is given and network weights  $w$  optimized
- Could also freeze weights  $w$  and optimize  $x$ , adversarially!

$$\min_{\delta} \text{size}(\delta) \quad \text{s.t.} \quad \text{pred}[f(w; x + \delta)] \neq y$$

or

$$\max_{\delta} l(w; x + \delta, y) \quad \text{s.t.} \quad \text{size}(\delta) \leq \epsilon, \quad 0 \leq x + \delta \leq 1$$

# Idea of Projected Gradient Descent

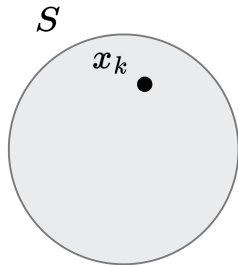


Figure 2: Suppose, we start from a point  $x_k$ .

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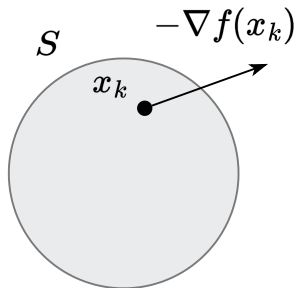


Figure 3: And go in the direction of  $-\nabla f(x_k)$ .

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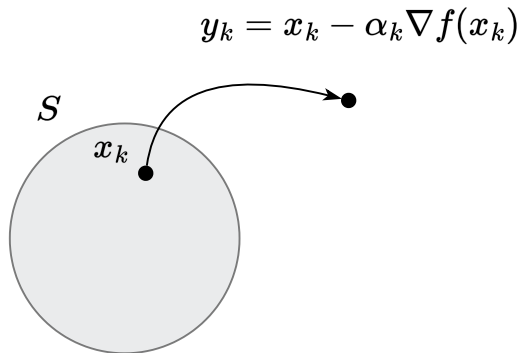


Figure 4: Occasionally, we can end up outside the feasible set.

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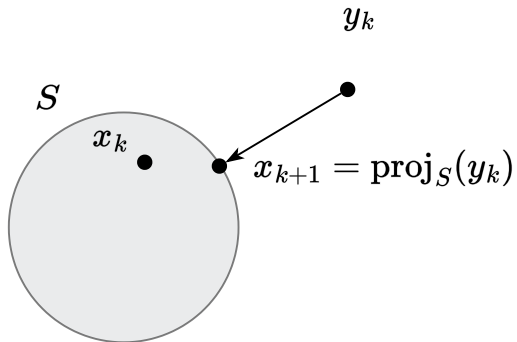


Figure 5: Solve this little problem with projection!

# Idea of Projected Gradient Descent

$$x_{k+1} = \text{proj}_S(x_k - \alpha_k \nabla f(x_k)) \quad \Leftrightarrow \quad \begin{aligned} y_k &= x_k - \alpha_k \nabla f(x_k) \\ x_{k+1} &= \text{proj}_S(y_k) \end{aligned}$$

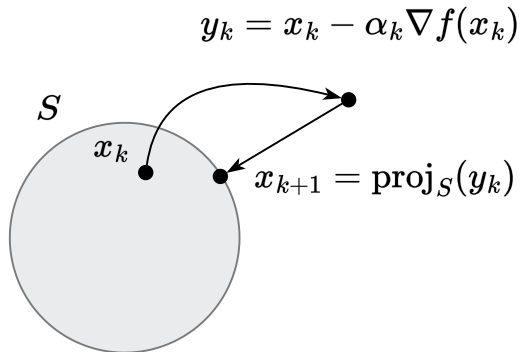


Figure 6: Illustration of Projected Gradient Descent algorithm

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# Projection criterion (Bourbaki-Cheney-Goldstein inequality)

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Let  $S \subseteq \mathbb{R}^n$  be closed and convex,  $\forall x \in S, y \in \mathbb{R}^n$ . Then

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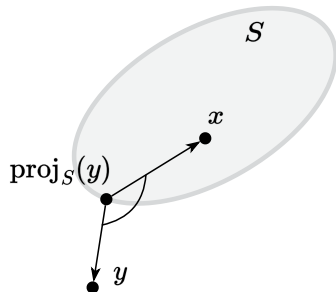


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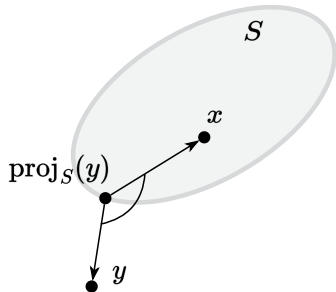


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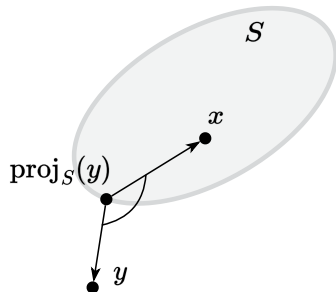


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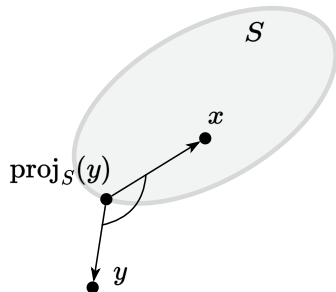


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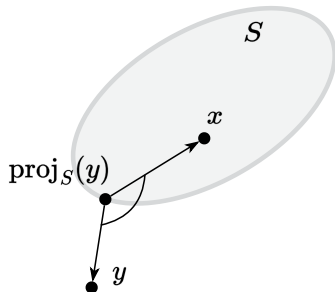


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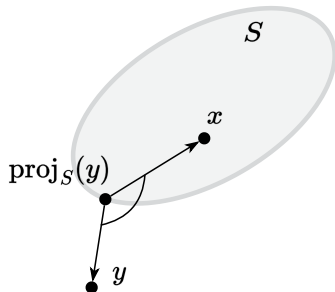


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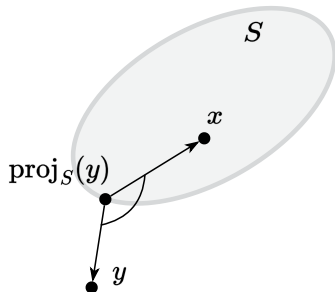


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It means the distance between the mapped points is possibly smaller than that of the unmapped points.

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- Next: variational characterization implies non-expansiveness. i.e.,

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Begins with the variational characterization / obtuse angle inequality

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Replace  $x$  by  $\pi(x)$  in Equation 3

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(Equation 4)+(Equation 5) will cancel  $\pi(y) - \pi(x)$ , not good. So flip the sign of (Equation 5) gives

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By Cauchy-Schwarz inequality, the left-hand-side is upper bounded by  $\|y - x\|_2 \|\pi(y) - \pi(x)\|_2$ , we get  $\|y - x\|_2 \|\pi(y) - \pi(x)\|_2 \geq \|\pi(x) - \pi(y)\|_2^2$ . Cancels  $\|\pi(x) - \pi(y)\|_2$  finishes the proof.

## Example: projection on the ball

Find  $\pi_S(y) = \pi$ , if  $S = \{x \in \mathbb{R}^n \mid \|x - x_0\| \leq R\}$ ,  $y \notin S$

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$$\begin{aligned} & \left( x_0 - y + R \frac{y - x_0}{\|y - x_0\|} \right)^T \left( x - x_0 - R \frac{y - x_0}{\|y - x_0\|} \right) = \\ & \left( \frac{(y - x_0)(R - \|y - x_0\|)}{\|y - x_0\|} \right)^T \left( \frac{(x - x_0)\|y - x_0\| - R(y - x_0)}{\|y - x_0\|} \right) = \\ & \frac{R - \|y - x_0\|}{\|y - x_0\|^2} (y - x_0)^T ((x - x_0)\|y - x_0\| - R(y - x_0)) = \\ & \frac{R - \|y - x_0\|}{\|y - x_0\|} \left( (y - x_0)^T (x - x_0) - R\|y - x_0\| \right) = \\ & (R - \|y - x_0\|) \left( \frac{(y - x_0)^T (x - x_0)}{\|y - x_0\|} - R \right) \end{aligned}$$

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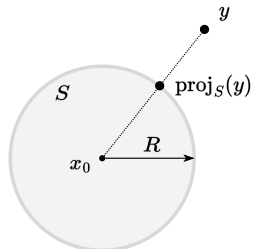
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## Example: projection on the halfspace

Find  $\pi_S(y) = \pi$ , if  $S = \{x \in \mathbb{R}^n \mid c^T x = b\}$ ,  $y \notin S$ . Build a hypothesis from the figure:  $\pi = y + \alpha c$ . Coefficient  $\alpha$  is chosen so that  $\pi \in S$ :  $c^T \pi = b$ , so:

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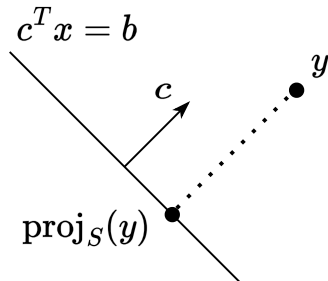


Figure 9: Hyperplane

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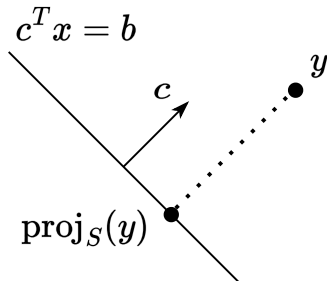


Figure 9: Hyperplane

$$c^T(y + \alpha c) = b$$

$$c^T y + \alpha c^T c = b$$

$$c^T y = b - \alpha c^T c$$

Check the inequality for a convex closed set:

$$(\pi - y)^T(x - \pi) \geq 0$$

$$(y + \alpha c - y)^T(x - y - \alpha c) =$$

$$\alpha c^T(x - y - \alpha c) =$$

$$\alpha(c^T x) - \alpha(c^T y) - \alpha^2(c^T c) =$$

$$\alpha b - \alpha(b - \alpha c^T c) - \alpha^2 c^T c =$$

$$\alpha b - \alpha b + \alpha^2 c^T c - \alpha^2 c^T c = 0 \geq 0$$

## Projected Gradient Descent (PGD)

## Idea

$$x_{k+1} = \text{proj}_S(x_k - \alpha_k \nabla f(x_k)) \quad \Leftrightarrow \quad \begin{aligned} y_k &= x_k - \alpha_k \nabla f(x_k) \\ x_{k+1} &= \text{proj}_S(y_k) \end{aligned}$$

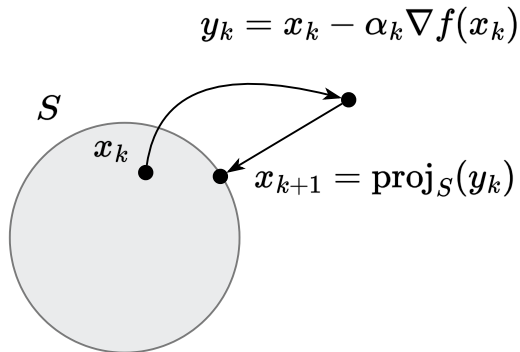


Figure 10: Illustration of Projected Gradient Descent algorithm

## Convergence rate for smooth and convex case

### Theorem

Let  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  be convex and differentiable. Let  $S \subseteq \mathbb{R}^n$  be a closed convex set, and assume that there is a minimizer  $x^*$  of  $f$  over  $S$ ; furthermore, suppose that  $f$  is smooth over  $S$  with parameter  $L$ . The Projected Gradient Descent algorithm with stepsize  $\frac{1}{L}$  achieves the following convergence after iteration  $k > 0$ :

$$f(x_k) - f^* \leq \frac{L\|x_0 - x^*\|_2^2}{2k}$$

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$$\begin{aligned} &= f(x_k) - \frac{L}{2}(\|y_k - x_k\|^2 + \|x_{k+1} - x_k\|^2 - \|y_k - x_{k+1}\|^2) + \frac{L}{2}\|x_{k+1} - x_k\|^2 \\ &= f(x_k) - \frac{1}{2L}\|\nabla f(x_k)\|^2 + \frac{L}{2}\|y_k - x_{k+1}\|^2 \end{aligned} \quad (7)$$

## Convergence rate for smooth and convex case

2. Now we do not immediately have progress at each step. Let's use again cosine rule:

$$\left\langle \frac{1}{L} \nabla f(x_k), x_k - x^* \right\rangle = \frac{1}{2} \left( \frac{1}{L^2} \|\nabla f(x_k)\|^2 + \|x_k - x^*\|^2 - \|x_k - x^* - \frac{1}{L} \nabla f(x_k)\|^2 \right)$$
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## Convergence rate for smooth and convex case

2. Now we do not immediately have progress at each step. Let's use again cosine rule:

$$\begin{aligned}\left\langle \frac{1}{L} \nabla f(x_k), x_k - x^* \right\rangle &= \frac{1}{2} \left( \frac{1}{L^2} \|\nabla f(x_k)\|^2 + \|x_k - x^*\|^2 - \|x_k - x^* - \frac{1}{L} \nabla f(x_k)\|^2 \right) \\ \langle \nabla f(x_k), x_k - x^* \rangle &= \frac{L}{2} \left( \frac{1}{L^2} \|\nabla f(x_k)\|^2 + \|x_k - x^*\|^2 - \|y_k - x^*\|^2 \right)\end{aligned}$$

3. We will use now projection property:  $\|x - \text{proj}_S(y)\|^2 + \|y - \text{proj}_S(y)\|^2 \leq \|x - y\|^2$  with  $x = x^*, y = y_k$ :

$$\begin{aligned}\|x^* - \text{proj}_S(y_k)\|^2 + \|y_k - \text{proj}_S(y_k)\|^2 &\leq \|x^* - y_k\|^2 \\ \|y_k - x^*\|^2 &\geq \|x^* - x_{k+1}\|^2 + \|y_k - x_{k+1}\|^2\end{aligned}$$

4. Now, using convexity and previous part:

$$\begin{aligned}\text{Convexity:} \quad f(x_k) - f^* &\leq \langle \nabla f(x_k), x_k - x^* \rangle \\ &\leq \frac{L}{2} \left( \frac{1}{L^2} \|\nabla f(x_k)\|^2 + \|x_k - x^*\|^2 - \|x_{k+1} - x^*\|^2 - \|y_k - x_{k+1}\|^2 \right)\end{aligned}$$

$$\text{Sum for } i = 0, k-1 \quad \sum_{i=0}^{k-1} [f(x_i) - f^*] \leq \sum_{i=0}^{k-1} \frac{1}{2L} \|\nabla f(x_i)\|^2 + \frac{L}{2} \|x_0 - x^*\|^2 - \frac{L}{2} \sum_{i=0}^{i-1} \|y_i - x_{i+1}\|^2$$

## Convergence rate for smooth and convex case

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## Convergence rate for smooth and convex case

6. From the sufficient decrease inequality

$$f(x_{k+1}) \leq f(x_k) - \frac{1}{2L} \|\nabla f(x_k)\|^2 + \frac{L}{2} \|y_k - x_{k+1}\|^2,$$

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Substitute back into (\*):

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Hence

$$f(x_{k+1}) \leq f(x_k) \quad \text{for each } k,$$

so  $\{f(x_k)\}$  is a monotonically nonincreasing sequence.

## Convergence rate for smooth and convex case

7. Final convergence bound From step 5, we have already established

$$\sum_{i=0}^{k-1} [f(x_i) - f^*] \leq \frac{L}{2} \|x_0 - x^*\|_2^2.$$

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which immediately gives

$$f(x_k) - f^* \leq \frac{L \|x_0 - x^*\|_2^2}{2k}.$$

This completes the proof of the  $\mathcal{O}(\frac{1}{k})$  convergence rate for convex and  $L$ -smooth  $f$  under projection constraints.

## Convergence rate for smooth strongly convex case

### Theorem

Let  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  be  $\mu$ -strongly convex (or, PL-function with parameter  $\mu$ ) and  $L$ -smooth. Let  $S \subseteq \mathbb{R}^n$  be a closed convex set, and assume that there is a minimizer  $x^*$  of  $f$  over  $S$ ; The Projected Gradient Descent algorithm with stepsize  $\frac{1}{L}$  achieves the following convergence after iteration  $k > 0$ :

$$f(x_k) - f^* \leq \left(1 - \frac{\mu}{L}\right)^k (f(x_0) - f^*)$$

1. We will use the notation  $x_{k+1} = \text{proj}_S(x_k - \frac{1}{L}\nabla f(x_k)) = \pi(y_k)$ . Since projection is nonexpansive, we have:

$$\|x_{k+1} - x_k\| = \|\pi(y_k) - \pi(x_k)\| \leq \|y_k - x_k\| = \frac{1}{L}\|\nabla f(x_k)\| \quad \Rightarrow \quad \|x_{k+1} - x_k\|^2 \leq \frac{1}{L^2}\|\nabla f(x_k)\|^2 \quad (8)$$

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## Convergence rate for smooth strongly convex case

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## Convergence rate for smooth strongly convex case

3. By first-order optimality for that Euclidean projection problem,  $\langle y_k - x_{k+1}, z - x_{k+1} \rangle \leq 0 \quad \forall z \in S$ . In particular, for  $z = x_k$ ,

$$\begin{aligned}\langle y_k - x_{k+1}, x_k - x_{k+1} \rangle &\leq 0 \\ \langle x_k - \frac{1}{L} \nabla f(x_k) - x_{k+1}, x_k - x_{k+1} \rangle &\leq 0 \\ \langle x_k - x_{k+1}, x_k - x_{k+1} \rangle - \frac{1}{L} \langle \nabla f(x_k), x_k - x_{k+1} \rangle &\leq 0 \\ \langle \nabla f(x_k), x_{k+1} - x_k \rangle &\leq -L \|x_{k+1} - x_k\|^2 \\ \langle \nabla f(x_k), x_{k+1} - x_k \rangle &\leq -L \frac{1}{L^2} \|\nabla f(x_k)\|^2 = -\frac{1}{L} \|\nabla f(x_k)\|^2\end{aligned}$$

4. Returning back to the smoothness inequality (9) we obtain sufficient decrease property:

$$\begin{aligned}f(x_{k+1}) &\leq f(x_k) + \langle \nabla f(x_k), x_{k+1} - x_k \rangle + \frac{1}{2L} \|\nabla f(x_k)\|^2 \\ &\leq f(x_k) - \frac{1}{L} \|\nabla f(x_k)\|^2 + \frac{1}{2L} \|\nabla f(x_k)\|^2 \\ &\leq f(x_k) - \frac{1}{2L} \|\nabla f(x_k)\|^2\end{aligned}$$

## Convergence rate for smooth strongly convex case

5. If we have a PL-function, we can use the following inequality:  $\|\nabla f(x_k)\|^2 \geq 2\mu (f(x_k) - f^*)$ . Thus, we can write:

$$\begin{aligned} f(x_{k+1}) &\leq f(x_k) - \frac{\mu}{L} (f(x_k) - f^*) \\ f(x_{k+1}) - f^* &\leq f(x_k) - f^* - \frac{\mu}{L} (f(x_k) - f^*) \end{aligned}$$

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$$\begin{aligned} f(x_{k+1}) &\leq f(x_k) - \frac{\mu}{L} (f(x_k) - f^*) \\ f(x_{k+1}) - f^* &\leq f(x_k) - f^* - \frac{\mu}{L} (f(x_k) - f^*) \end{aligned}$$

6. Now we can use induction to get the following bound:

$$f(x_k) - f^* \leq \left(1 - \frac{\mu}{L}\right)^k (f(x_0) - f^*)$$

## Frank-Wolfe Method



Figure 11: Marguerite Straus Frank (1927-2024)



Figure 12: Philip Wolfe (1927-2016)

# Idea

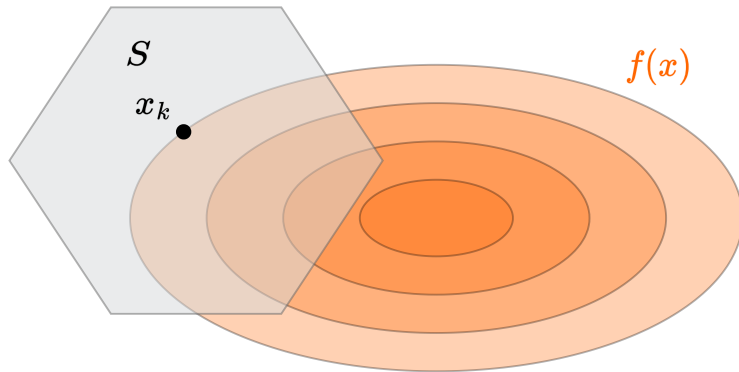


Figure 13: Illustration of Frank-Wolfe (conditional gradient) algorithm

# Idea

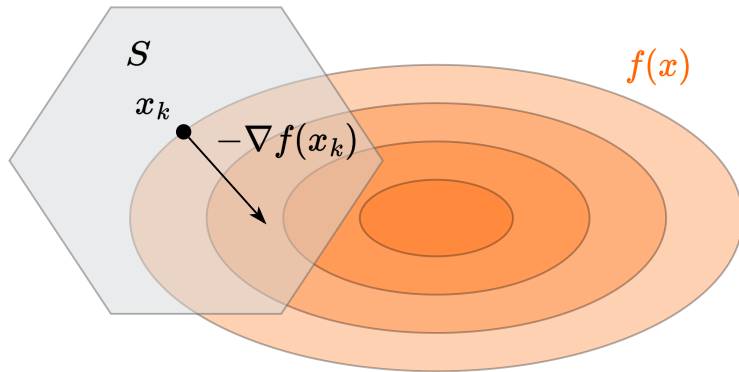


Figure 14: Illustration of Frank-Wolfe (conditional gradient) algorithm

# Idea

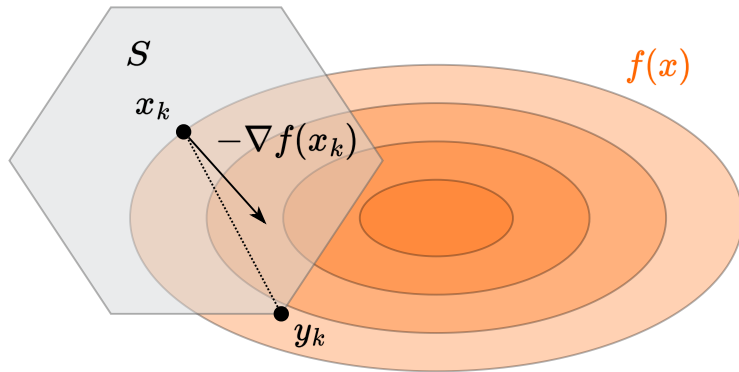


Figure 15: Illustration of Frank-Wolfe (conditional gradient) algorithm

# Idea

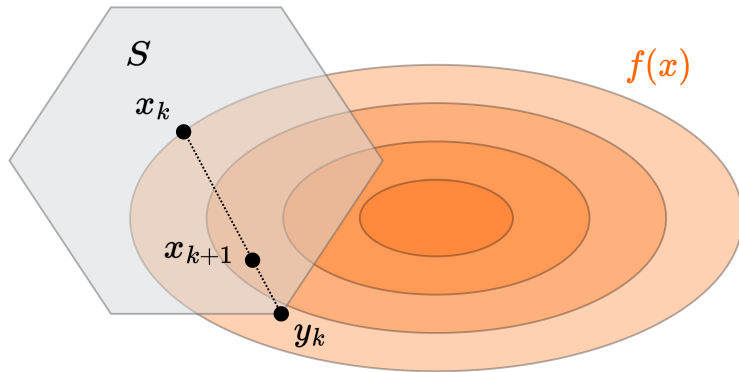


Figure 16: Illustration of Frank-Wolfe (conditional gradient) algorithm

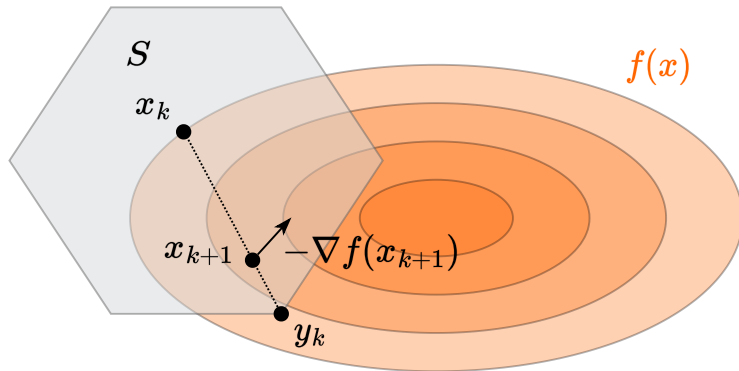


Figure 17: Illustration of Frank-Wolfe (conditional gradient) algorithm

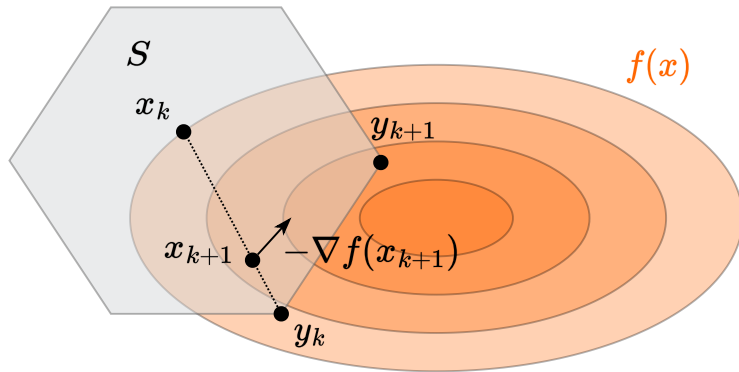


Figure 18: Illustration of Frank-Wolfe (conditional gradient) algorithm

# Idea

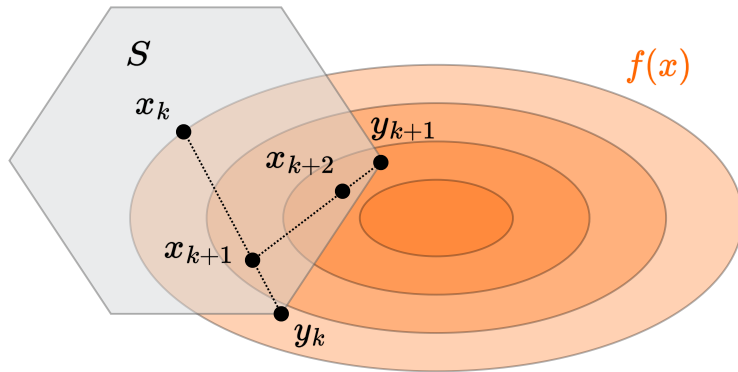


Figure 19: Illustration of Frank-Wolfe (conditional gradient) algorithm

## Idea

$$y_k = \arg \min_{x \in S} f_{x_k}^I(x) = \arg \min_{x \in S} \langle \nabla f(x_k), x \rangle$$

$$x_{k+1} = \gamma_k x_k + (1 - \gamma_k) y_k$$

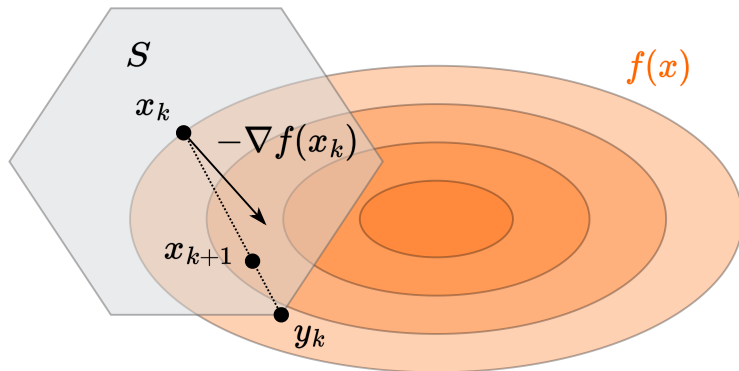


Figure 20: Illustration of Frank-Wolfe (conditional gradient) algorithm

## Convergence rate for smooth and convex case

### Theorem

Let  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  be convex and differentiable. Let  $S \subseteq \mathbb{R}^n$  be a closed convex set, and assume that there is a minimizer  $x^*$  of  $f$  over  $S$ ; furthermore, suppose that  $f$  is smooth over  $S$  with parameter  $L$ . The Frank-Wolfe algorithm with step size  $\gamma_k = \frac{k-1}{k+1}$  achieves the following convergence after iteration  $k > 0$ :

$$f(x_k) - f^* \leq \frac{2LR^2}{k+1}$$

where  $R = \max_{x,y \in S} \|x - y\|$  is the diameter of the set  $S$ .

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1. By  $L$ -smoothness of  $f$ , we have:

$$\begin{aligned} f(x_{k+1}) - f(x_k) &\leq \langle \nabla f(x_k), x_{k+1} - x_k \rangle + \frac{L}{2} \|x_{k+1} - x_k\|^2 \\ &= (1 - \gamma_k) \langle \nabla f(x_k), y_k - x_k \rangle + \frac{L(1 - \gamma_k)^2}{2} \|y_k - x_k\|^2 \end{aligned}$$

## Convergence rate for smooth and convex case

2. By convexity of  $f$ , for any  $x \in S$ , including  $x^*$ :

$$\langle \nabla f(x_k), x - x_k \rangle \leq f(x) - f(x_k)$$

In particular, for  $x = x^*$ :

$$\langle \nabla f(x_k), x^* - x_k \rangle \leq f(x^*) - f(x_k)$$

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3. By definition of  $y_k$ , we have  $\langle \nabla f(x_k), y_k \rangle \leq \langle \nabla f(x_k), x^* \rangle$ , thus:

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4. Combining the above inequalities:

$$\begin{aligned} f(x_{k+1}) - f(x_k) &\leq (1 - \gamma_k) \langle \nabla f(x_k), y_k - x_k \rangle + \frac{L(1 - \gamma_k)^2}{2} \|y_k - x_k\|^2 \\ &\leq (1 - \gamma_k) (f(x^*) - f(x_k)) + \frac{L(1 - \gamma_k)^2}{2} R^2 \end{aligned}$$

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5. Rearranging terms:

$$f(x_{k+1}) - f(x^*) \leq \gamma_k (f(x_k) - f(x^*)) + (1 - \gamma_k)^2 \frac{LR^2}{2}$$

## Convergence rate for smooth and convex case

6. Denoting  $\delta_k = \frac{f(x_k) - f(x^*)}{LR^2}$ , we get:

$$\delta_{k+1} \leq \gamma_k \delta_k + \frac{(1 - \gamma_k)^2}{2} = \frac{k-1}{k+1} \delta_k + \frac{2}{(k+1)^2}$$

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- Then  $\delta_{k+1} \leq \frac{k-1}{k+1} \cdot \frac{2}{k+1} + \frac{2}{(k+1)^2} = \frac{2k}{k^2+2k+1} < \frac{2}{k+2}$  

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## Lower bound for Frank-Wolfe method <sup>2</sup>

### Theorem

Consider any algorithm that accesses the feasible set  $S \subseteq \mathbb{R}^n$  only via a linear minimization oracle (LMO). Let the diameter of the set  $S$  be  $R$ . There exists an  $L$ -smooth strongly convex function  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  such that this algorithm requires at least

$$\min \left( \frac{n}{2}, \frac{LR^2}{16\varepsilon} \right)$$

iterations (i.e., calls to the LMO) to construct a point  $\hat{x} \in S$  with  $f(\hat{x}) - \min_{x \in S} f(x) \leq \varepsilon$ . The lower bound applies both for convex and strongly convex functions.

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<sup>2</sup>  The Complexity of Large-scale Convex Programming under a Linear Optimization Oracle

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**Sketch of the proof.** Consider the following optimization problem: Note, that:

- $f$  is 1-smooth;
- the diameter of  $S$  is  $R = 2$ ;
- $f$  is strongly convex.

$$\min_{x \in S} f(x) = \min_{x \in S} \frac{1}{2} \|x\|_2^2$$

$$S = \left\{ x \in \mathbb{R}^n \mid x \geq 0, \sum_{i=1}^n x_i = 1 \right\}$$

<sup>2</sup>  The Complexity of Large-scale Convex Programming under a Linear Optimization Oracle

## Lower bound for Frank-Wolfe method <sup>3</sup>

1. The optimal solution is

$$x^* = \frac{1}{n} \mathbf{1} = \frac{1}{n} \sum_{i=1}^n e_i, \quad \text{and} \quad f(x^*) = \frac{1}{2n},$$

where  $e_i = (0, \dots, 0, \underset{\text{position } i}{1}, 0, \dots, 0)^\top$  is the  $i$ -th standard basis vector.

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2. A linear minimization oracle (LMO) over  $S$  returns a vertex  $e_i$ . After  $k$  iterations, the method will have discovered at most  $k$  different basis vectors  $e_{i_1}, \dots, e_{i_k}$ . The best convex combination one can form is

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4. To ensure that  $f(\hat{x}) - f(x^*) \leq \varepsilon$ , it is necessary that (full proof is in the paper):

$$k \geq \min \left\{ \frac{n}{2}, \frac{1}{4\varepsilon} \right\} = \min \left\{ \frac{n}{2}, \frac{LR^2}{16\varepsilon} \right\}.$$

---

<sup>3</sup>  The Complexity of Large-scale Convex Programming under a Linear Optimization Oracle

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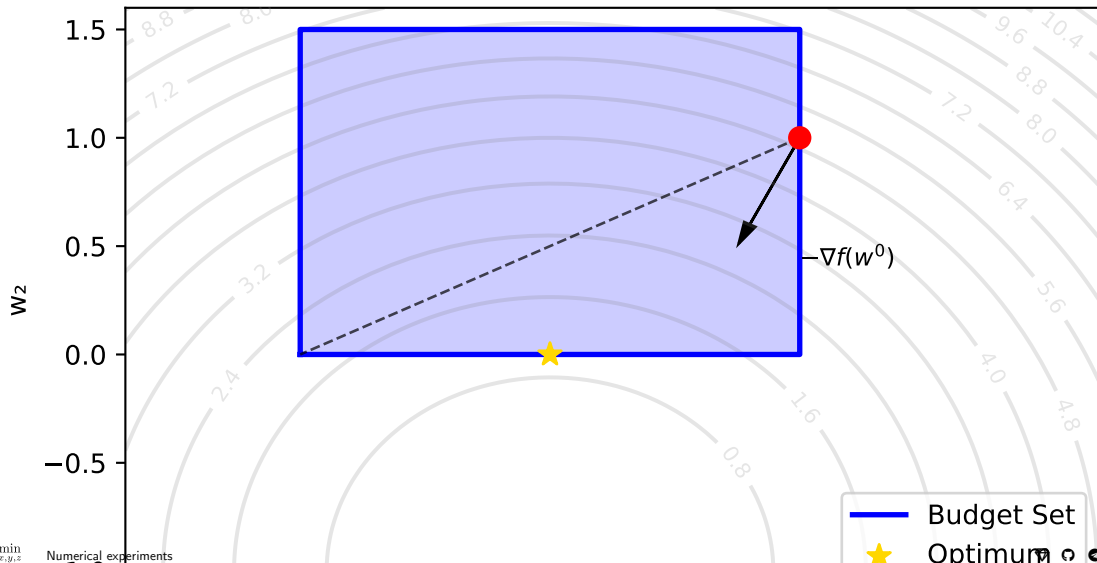
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- Recent work showed the extension to non-smooth case (📄 paper) with convergence rate  $O\left(\frac{1}{\sqrt{k}}\right)$

## Numerical experiments

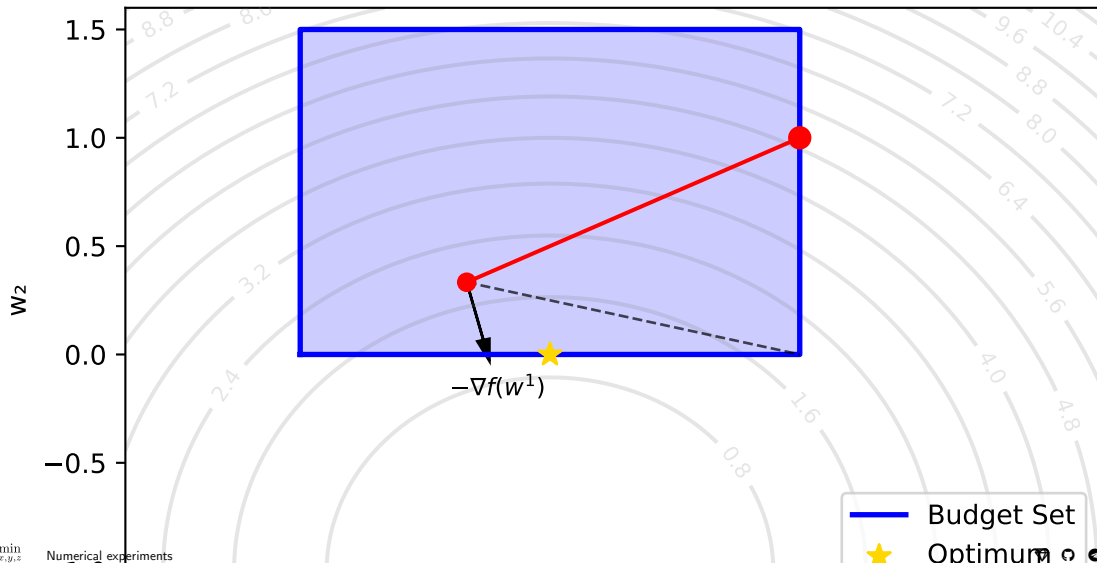
## 2d example. Frank-Wolfe method

Frank-Wolfe Method: Iteration 0



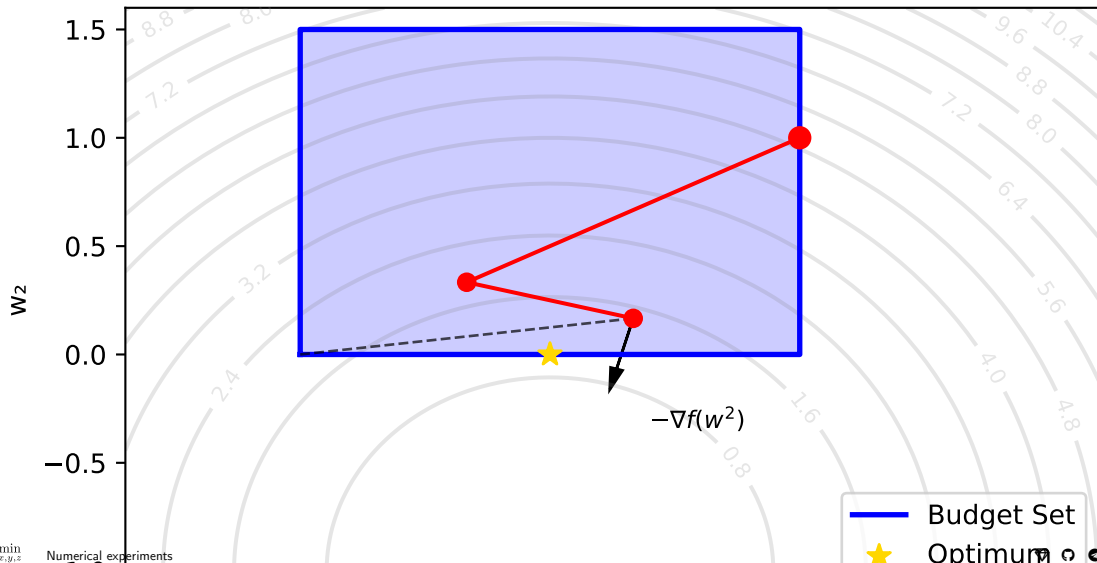
## 2d example. Frank-Wolfe method

Frank-Wolfe Method: Iteration 1



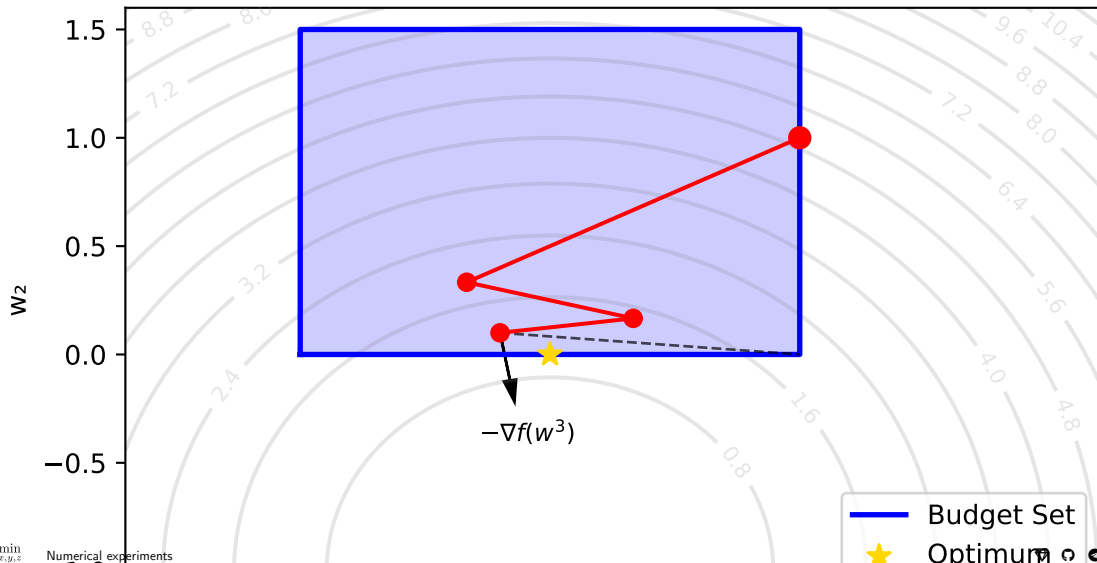
## 2d example. Frank-Wolfe method

Frank-Wolfe Method: Iteration 2



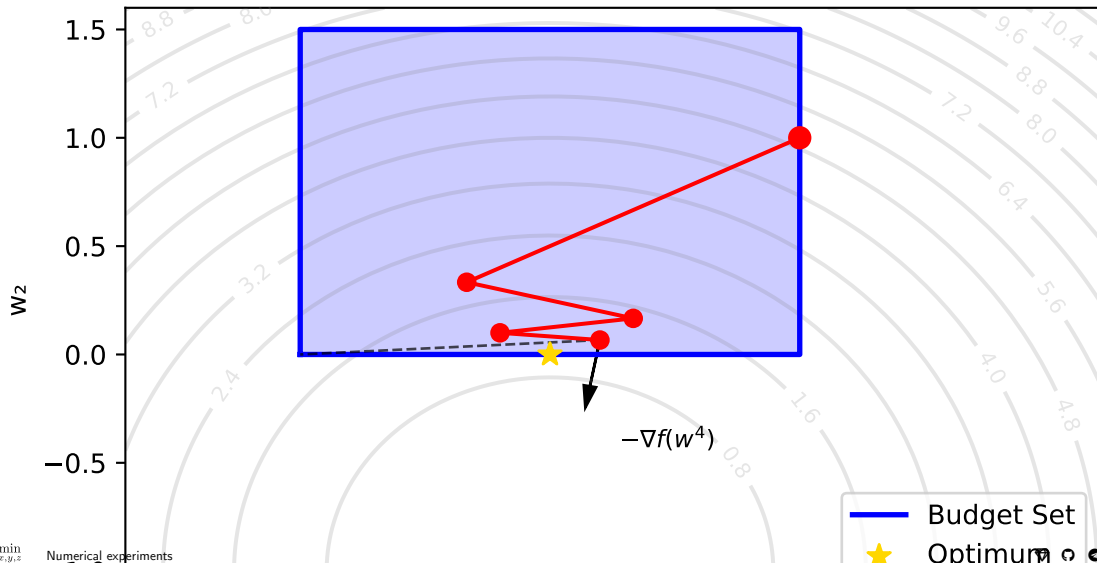
## 2d example. Frank-Wolfe method

Frank-Wolfe Method: Iteration 3



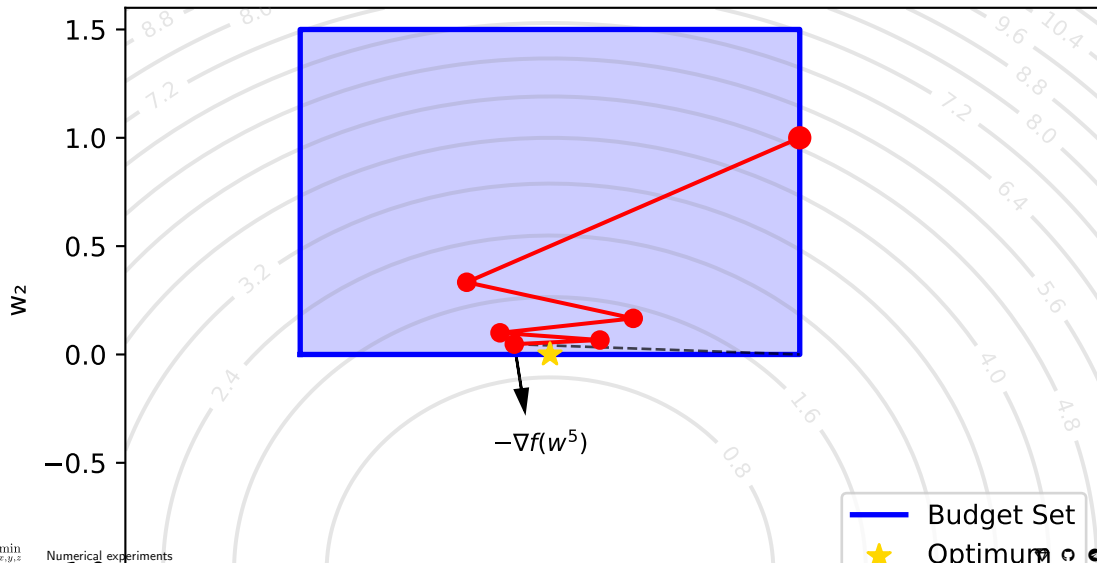
## 2d example. Frank-Wolfe method

Frank-Wolfe Method: Iteration 4



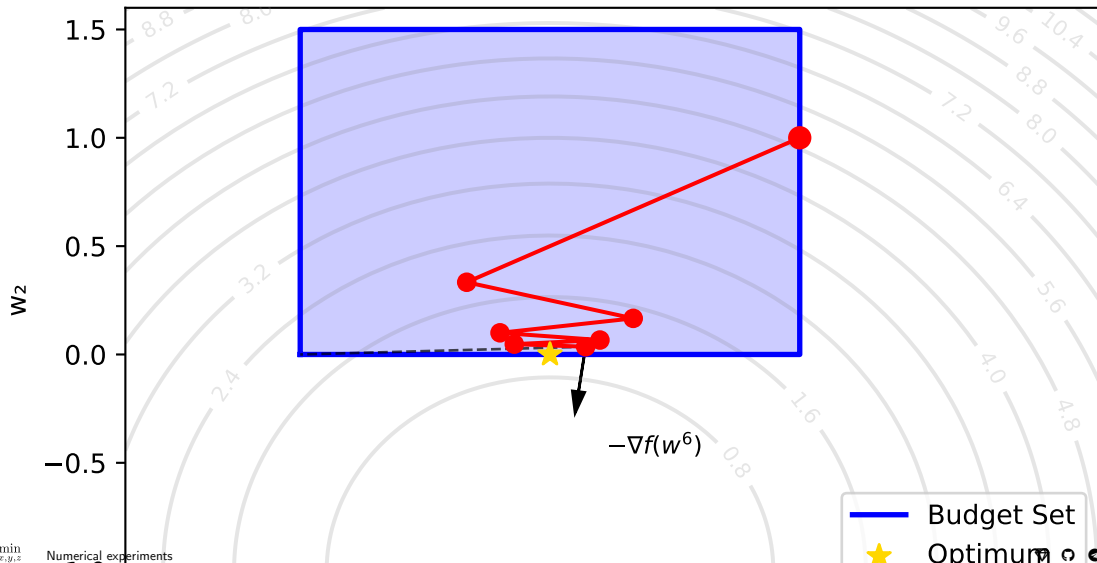
## 2d example. Frank-Wolfe method

Frank-Wolfe Method: Iteration 5



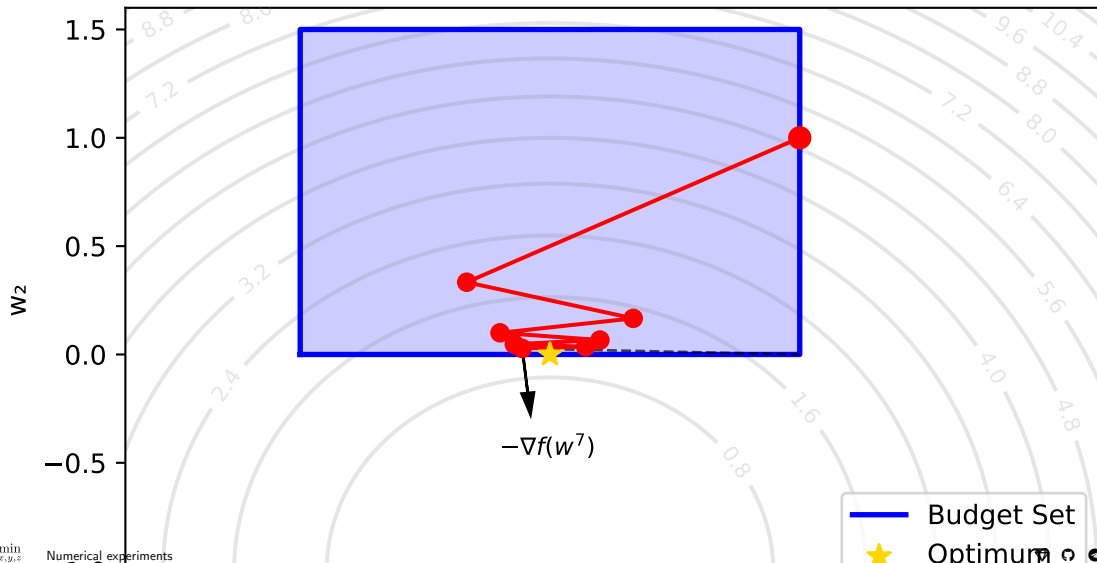
## 2d example. Frank-Wolfe method

Frank-Wolfe Method: Iteration 6



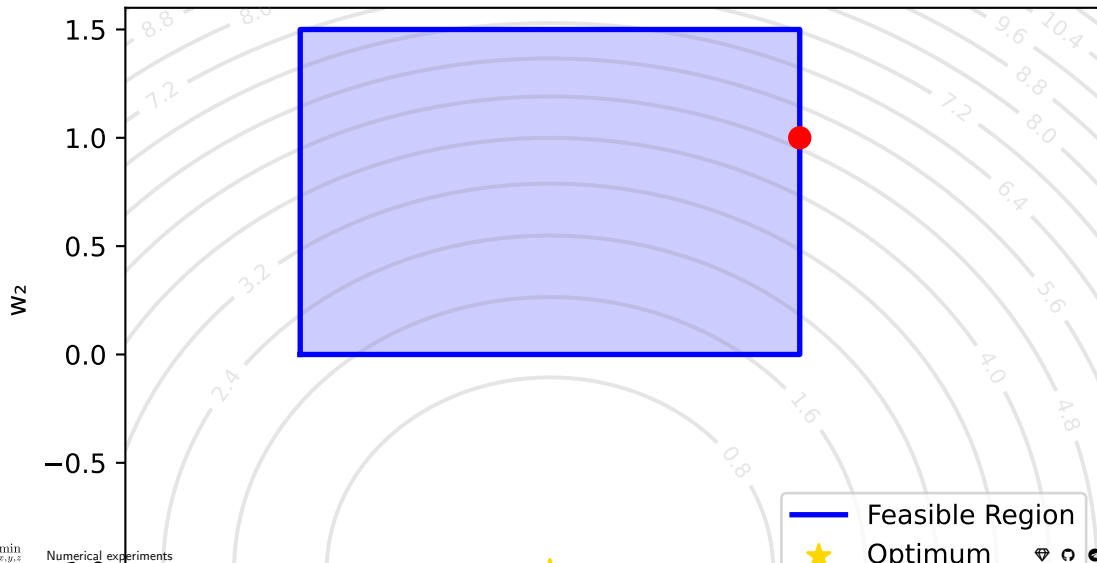
## 2d example. Frank-Wolfe method

Frank-Wolfe Method: Iteration 7



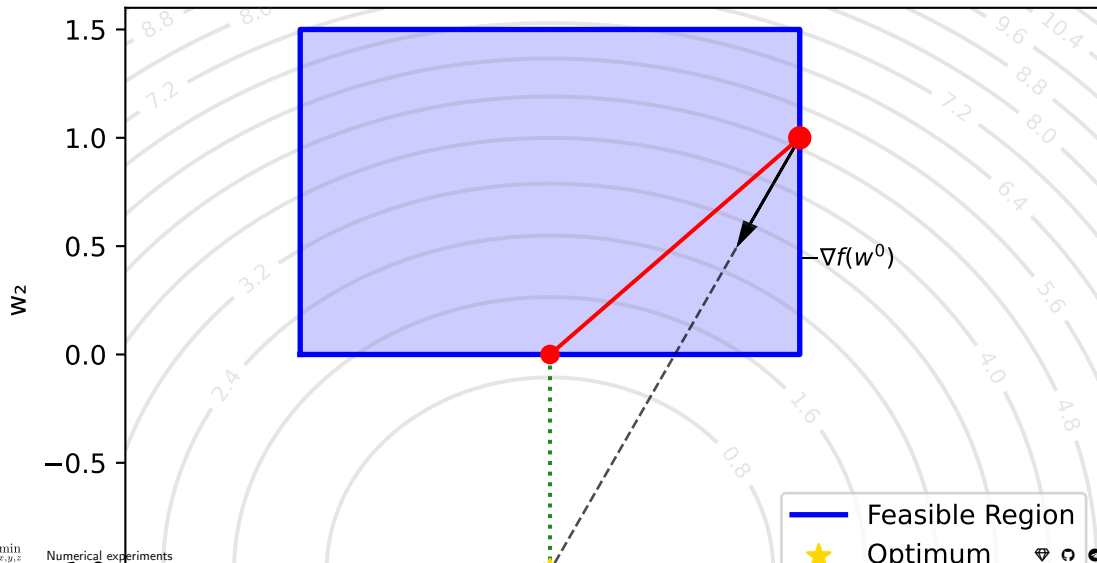
## 2d example. Projected gradient descent

Projected Gradient Descent: Iteration 0



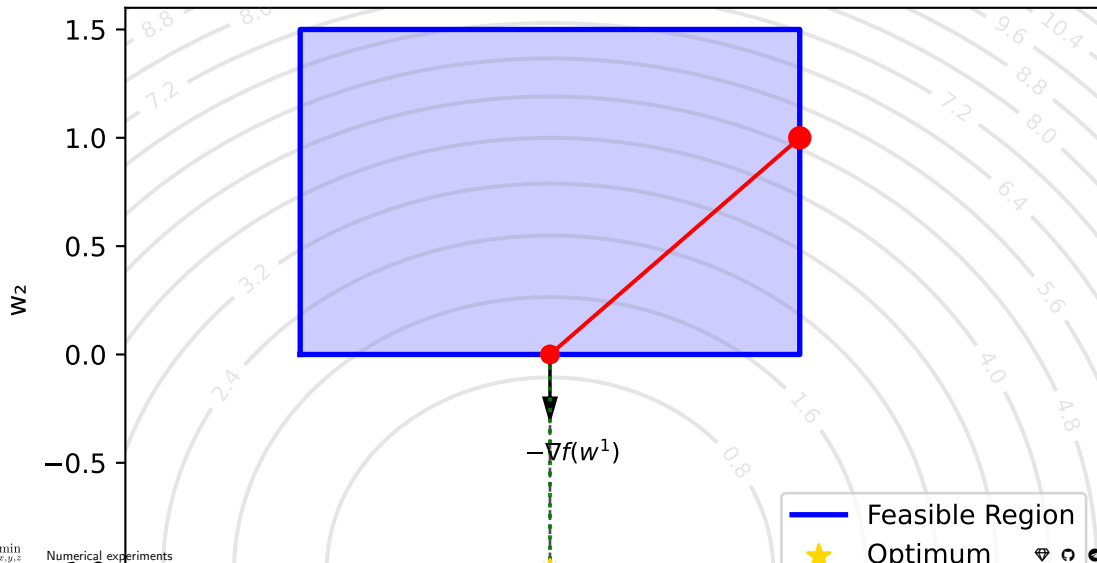
## 2d example. Projected gradient descent

Projected Gradient Descent: Iteration 1



## 2d example. Projected gradient descent

Projected Gradient Descent: Iteration 2



## Quadratic function. Box constraints

$$\min_{\substack{x \in \mathbb{R}^n \\ -1 \preceq x \preceq 1}} \frac{1}{2} x^\top A x - b^\top x,$$

$$A \in \mathbb{R}^{n \times n}, \quad \lambda(A) \in [\mu; L].$$

The projection is simple:

$$\pi_S(x) = \text{clip}(x, -1, 1).$$

or

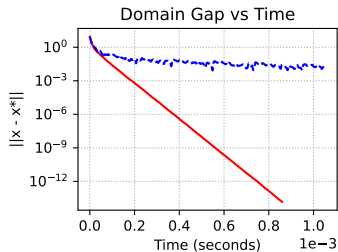
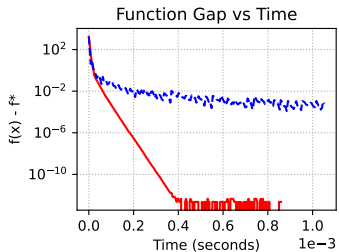
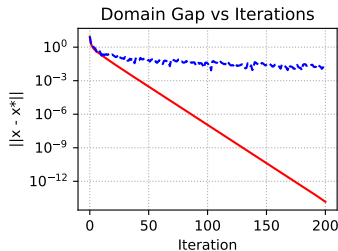
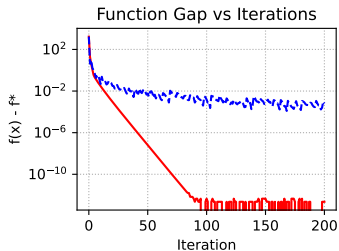
$$\pi_S(x) = \max(-1, \min(1, x)).$$

The linear minimization oracle (LMO) for a given gradient  $g$  is given by  $y = \underset{z \in S}{\operatorname{argmin}} \langle g, z \rangle$ .

Since the feasible set is separable across coordinates, the solution is computed coordinate-wise as

$$y_i = \begin{cases} -1, & \text{if } g_i > 0, \\ 1, & \text{if } g_i \leq 0. \end{cases}$$

Constrained convex quadratic problem:  $n=80, \mu=0, L=10$



— Projected Gradient Descent    - - - Frank-Wolfe

## Quadratic function. Box constraints

$$\min_{\substack{x \in \mathbb{R}^n \\ -1 \preceq x \preceq 1}} \frac{1}{2} x^\top A x - b^\top x,$$

$$A \in \mathbb{R}^{n \times n}, \quad \lambda(A) \in [\mu; L].$$

The projection is simple:

$$\pi_S(x) = \text{clip}(x, -1, 1).$$

or

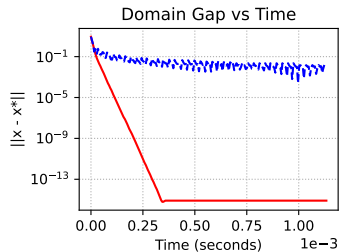
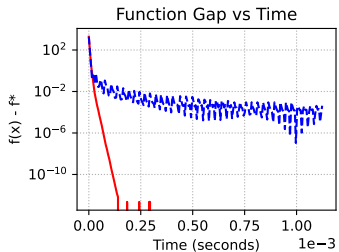
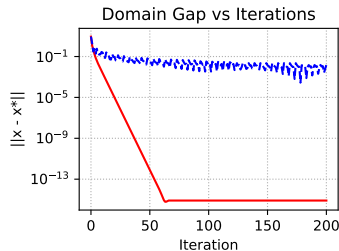
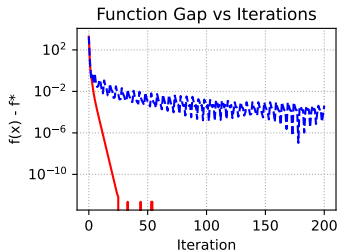
$$\pi_S(x) = \max(-1, \min(1, x)).$$

The linear minimization oracle (LMO) for a given gradient  $g$  is given by  $y = \underset{z \in S}{\operatorname{argmin}} \langle g, z \rangle$ .

Since the feasible set is separable across coordinates, the solution is computed coordinate-wise as

$$y_i = \begin{cases} -1, & \text{if } g_i > 0, \\ 1, & \text{if } g_i \leq 0. \end{cases}$$

Constrained strongly Convex quadratic problem:  $n=80, \mu=1, L=10$



— Projected Gradient Descent    - - - Frank-Wolfe

# Quadratic function. Simplex constraints (Lucky problem with diagonal matrix)

$$\min_{\substack{x \in \mathbb{R}^n \\ x \geq 0, 1^T x = 1}} \frac{1}{2} x^T A x,$$

$$A \in \mathbb{R}^{n \times n}, \quad \lambda(A) \in [0; 100].$$

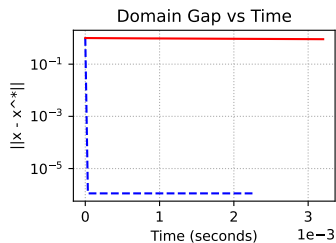
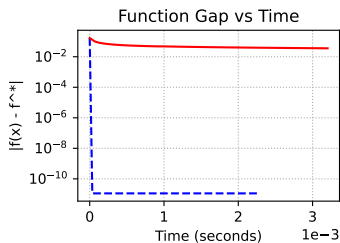
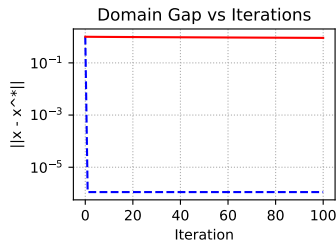
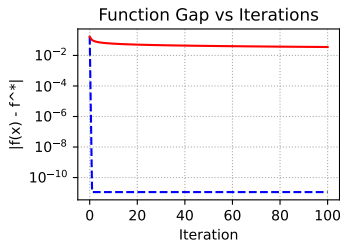
$$\min_{1^T x = 1, x \geq 0} \frac{1}{2} x^T A x, n = 200$$

| Method | Update time, ms | LMO/Projection |
|--------|-----------------|----------------|
| PGD    | 0.0069          | 0.0167         |
| FW     | 0.0070          | 0.0066         |

The projection onto the unit simplex  $\pi_S(x)$  can be done in  $\mathcal{O}(n \log n)$  or expected  $\mathcal{O}(n)$  time.<sup>a</sup>

The LMO for a given gradient  $g$  is given by  $y = \operatorname{argmin}_{z \in S} \langle g, z \rangle$ . The solution corresponds to a vertex of the simplex:

$$y = e_j \quad \text{where} \quad j = \operatorname{argmin}_i g_i.$$



--- Frank-Wolfe    — Projected Gradient Descent

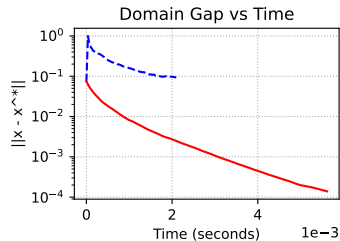
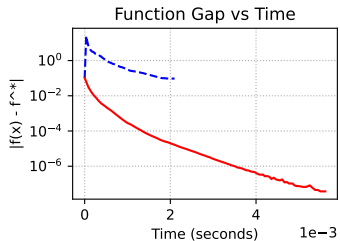
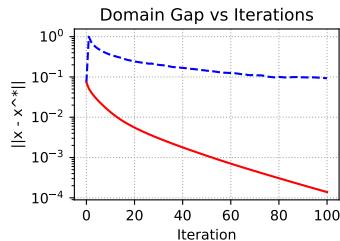
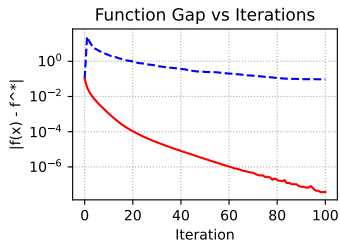
# Quadratic function. Simplex constraints

$$\min_{\substack{x \in \mathbb{R}^n \\ x \geq 0, 1^T x = 1}} \frac{1}{2} x^T A x,$$

$$A \in \mathbb{R}^{n \times n}, \quad \lambda(A) \in [0; 100].$$

$$\min_{1^T x = 1, x \geq 0} \frac{1}{2} x^T A x, n = 200$$

| Method | Update time, ms | LMO/Projection |
|--------|-----------------|----------------|
| PGD    | 0.0069          | 0.0420         |
| FW     | 0.0069          | 0.0066         |



-- Frank-Wolfe    — Projected Gradient Descent

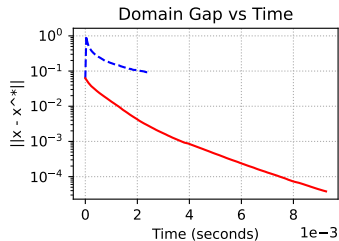
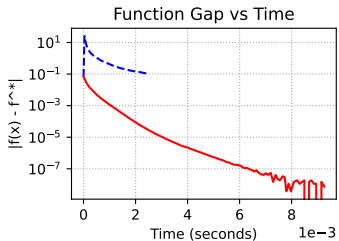
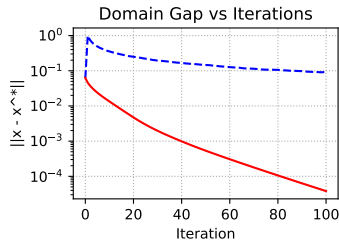
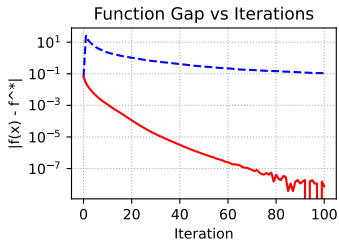
# Quadratic function. Simplex constraints

$$\min_{\substack{x \in \mathbb{R}^n \\ x \geq 0, 1^T x = 1}} \frac{1}{2} x^T A x,$$

$$A \in \mathbb{R}^{n \times n}, \quad \lambda(A) \in [0; 100].$$

$$\min_{1^T x = 1, x \geq 0} \frac{1}{2} x^T A x, n = 300$$

| Method | Update time, ms | LMO/Projection |
|--------|-----------------|----------------|
| PGD    | 0.0068          | 0.0761         |
| FW     | 0.0069          | 0.0070         |



-- Frank-Wolfe    — Projected Gradient Descent

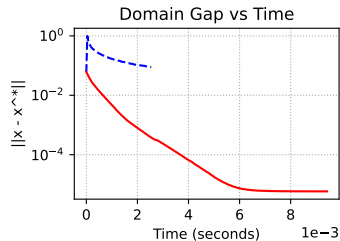
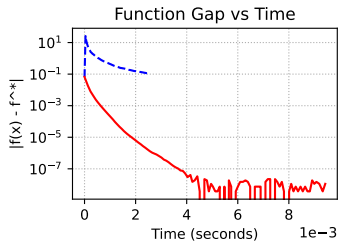
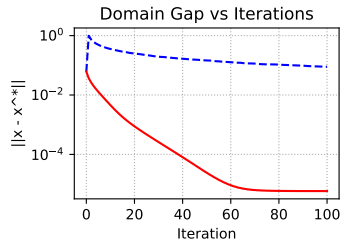
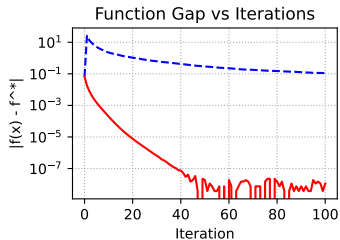
# Quadratic function. Simplex constraints

$$\min_{\substack{x \in \mathbb{R}^n \\ x \geq 0, 1^T x = 1}} \frac{1}{2} x^T A x,$$

$$A \in \mathbb{R}^{n \times n}, \quad \lambda(A) \in [1; 100].$$

$$\min_{1^T x = 1, x \geq 0} \frac{1}{2} x^T A x, n = 300$$

| Method | Update time, ms | LMO/Projection |
|--------|-----------------|----------------|
| PGD    | 0.0068          | 0.0752         |
| FW     | 0.0067          | 0.0068         |



-- Frank-Wolfe    — Projected Gradient Descent

## PGD vs Frank-Wolfe

The key difference between PGD and FW is that PGD requires projection, while FW needs only linear minimization oracle (LMO).

In a recent book authors presented the following comparison table with complexities of linear minimizations and projections on some convex sets up to an additive error  $\epsilon$  in the Euclidean norm.

| Set                                                                                | Linear minimization                                                            | Projection                                                                          |
|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| $n$ -dimensional $\ell_p$ -ball, $p \neq 1, 2, \infty$                             | $\mathcal{O}(n)$                                                               | $\tilde{\mathcal{O}}\left(\frac{n}{\epsilon^2}\right)$                              |
| Nuclear norm ball of $n \times m$ matrices                                         | $\mathcal{O}\left(\nu \ln(m+n) \frac{\sqrt{\sigma_1}}{\sqrt{\epsilon}}\right)$ | $\mathcal{O}(m n \min\{m, n\})$                                                     |
| Flow polytope on a graph with $m$ vertices and $n$ edges (capacity bound on edges) | $\mathcal{O}\left((n \log m)(n + m \log m)\right)$                             | $\tilde{\mathcal{O}}\left(\frac{n}{\epsilon^2}\right)$ or $\mathcal{O}(n^4 \log n)$ |
| Birkhoff polytope ( $n \times n$ doubly stochastic matrices)                       | $\mathcal{O}(n^3)$                                                             | $\tilde{\mathcal{O}}\left(\frac{n^2}{\epsilon^2}\right)$                            |

When  $\epsilon$  is missing, there is no additive error. The  $\tilde{\mathcal{O}}$  hides polylogarithmic factors in the dimensions and polynomial factors in constants related to the distance to the optimum. For the nuclear norm ball, i.e., the spectrahedron,  $\nu$  denotes the number of non-zero entries and  $\sigma_1$  denotes the top singular value of the projected matrix.